GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRE PRACTICAL TEST

Academic Period: JULY 2026

Program	**:**	**NETWORK SECURITY**
Course	**:**	**NETWORK MANAGEMENT & MONITORING SYSTEMS**
Code	**:**	**NTC1052**
Duration	**:**	**2 HOURS**

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SEM/TRADE/GROUP	**: SEM 5 / DCBS / CBS 1**
DATE	**: 13 / 10 / 2026**

ANSWER ALL QUESTIONS.

1. Install and configure Zabbix as your monitoring system. Ensure the web interface is accessible, and set up a user with read-only access for monitoring. Provide the steps to verify the Installation.

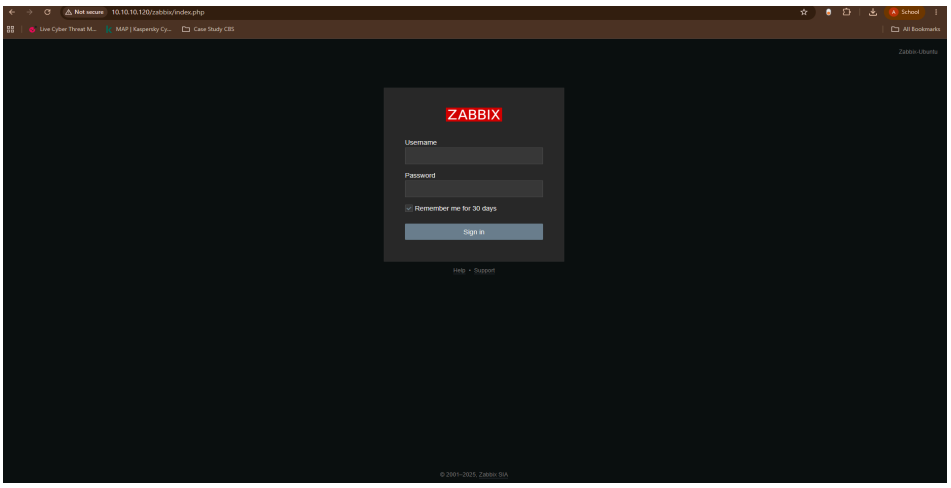
```
root@Ubuntu:~# systemctl status zabbix-server zabbix-agent apache2 mariadb --no-pager
● zabbix-server.service - Zabbix Server (MySQL/MariaDB)
     Loaded: loaded (/usr/lib/systemd/system/zabbix-server.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-10-13 16:55:25 +08; 50min ago
 Invocation: 38c1cfb7d87f43c3a928fc15c36b91c3
       Docs: man:zabbix_server
   Main PID: 20184 (zabbix_server)
      Tasks: 77 (limit: 3887)
     Memory: 60.4M (peak: 60.7M)
        CPU: 16.232s
```

```
● zabbix-agent.service - Zabbix Agent
     Loaded: loaded (/usr/lib/systemd/system/zabbix-agent.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-10-13 16:42:50 +08; 1h 3min ago
Invocation: dd3637dc158d48cfb36ec4e779c9c2fc
       Docs: man:zabbix_agentd
   Main PID: 17848 (zabbix_agentd)
      Tasks: 13 (limit: 3887)
     Memory: 8.9M (peak: 16.1M)
        CPU: 5.254s
     CGroup: /system.slice/zabbix-agent.service
```

```
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-10-13 16:44:39 +08; 1h 1min ago
Invocation: 385bc3f4d58648d2b5e204f4bd586742
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 19438 (apache2)
      Tasks: 11 (limit: 3887)
     Memory: 74.6M (peak: 85.2M)
        CPU: 13.898s
     CGroup: /system.slice/apache2.service
```

```
● mariadb.service - MariaDB 11.8.3 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-10-13 16:42:50 +08; 1h 3min ago
 Invocation: 1f5851d9cc684225a60c0760c968ed7d
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
   Main PID: 18004 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 50 (limit: 25654)
     Memory: 314.7M (peak: 315.7M)
        CPU: 16.733s
     CGroup: /system.slice/mariadb.service
             └─18004 /usr/sbin/mariadbd
```

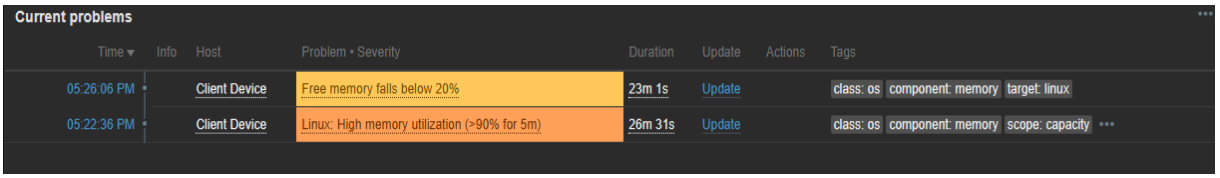

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2. Configure Zabbix to monitor two hosts:

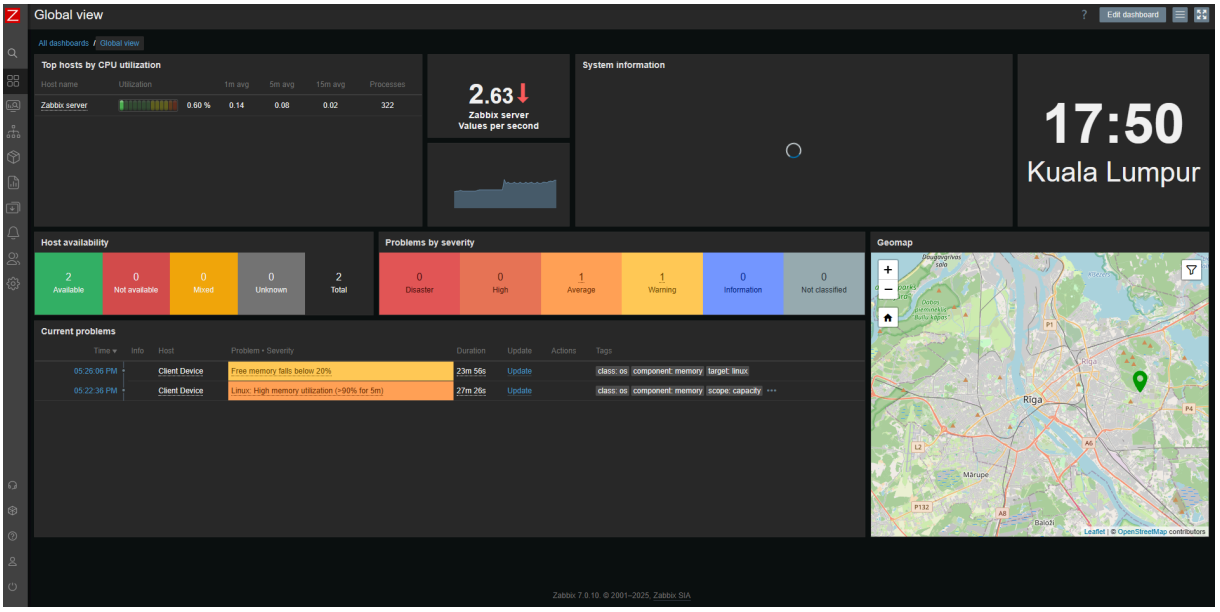
1) Your localhost with HTTP service and CPU load monitoring, setting custom trigger thresholds for high CPU usage (e.g., alert if CPU load exceeds 80% for 5 minutes).

2) A neighboring host with a ping service and memory usage monitoring, configuring a trigger to alert if free memory falls below 20%.

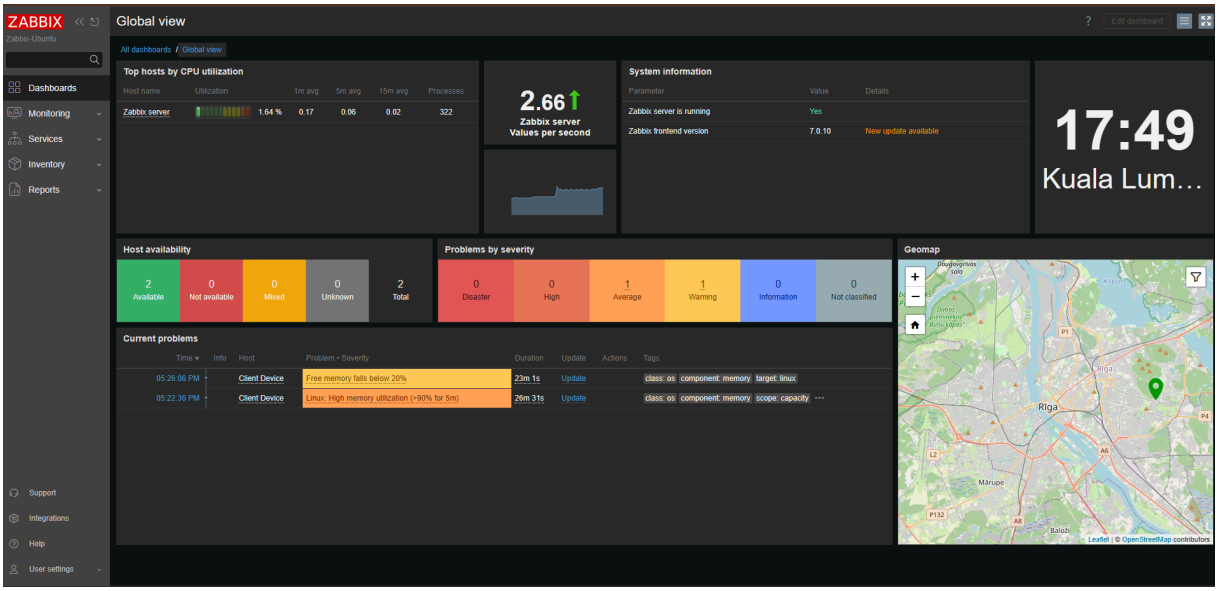


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On Admin dashboard :



On Monitor (Read-Only) dashboard :



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3. Answer the following questions, including screenshots where required:

1) Where is the main Zabbix configuration file located, and what is one key parameter you can modify to adjust the polling interval?

-> File Location is at /etc/zabbix/zabbix_server.conf (for the server daemon) and /etc/zabbix/zabbix_agentd.conf (for the agent).

2) Which configuration file or section in Zabbix is used to define new hosts, and what is a mandatory field for host configuration?

-> The section used to define new hosts is in the Data Collection dropdown -> Hosts. And the mandatory field is both the Host name and at least one Host group.

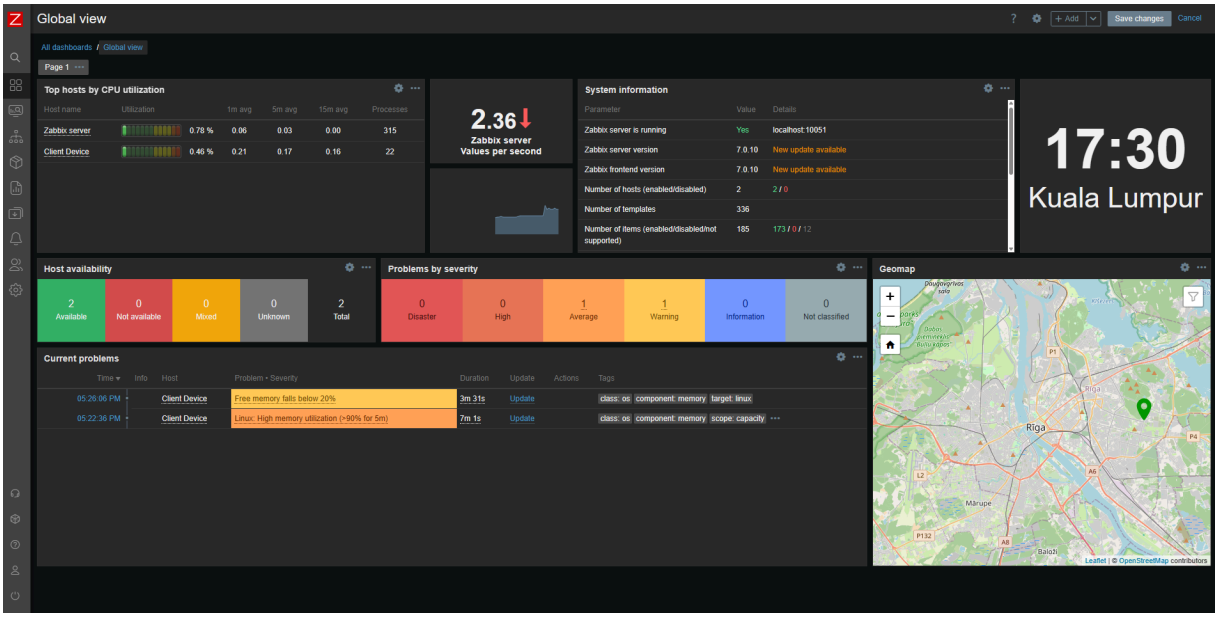
3) How do you configure Zabbix to send email notifications for triggered alerts? Specify the configuration section and one key setting.

-> Configure Zabbix to send email notifications for triggered alerts, is by opening Alerts dropdown menu and go to Media types (to configure SMTP transport) combined with Alerts -> Actions -> Trigger actions to create the alert dispatch rule.

-> Key Setting: Under Alerts -> Media types -> Email, configure the SMTP server (smtp.test.com / localhost) along with SMTP email and port.

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4) From the Zabbix dashboard, provide two pieces of information about the status of each monitored host (e.g., availability, latest trigger status). Include a screenshot.



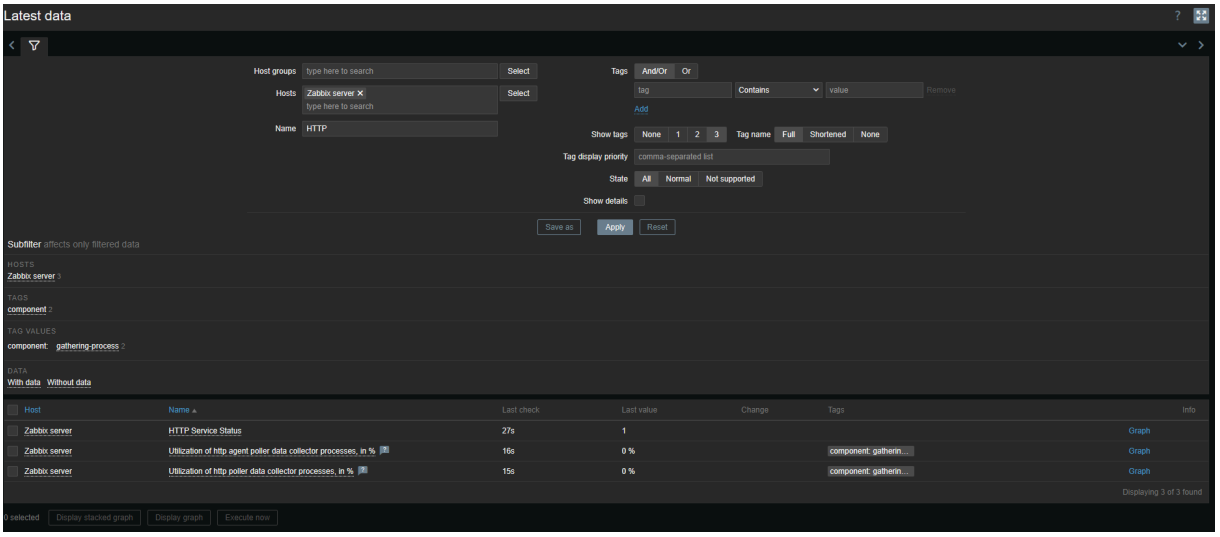
Based on the dashboard, both **Zabbix server** and **Client Device** show an **Available** status in the *Host availability* widget. In terms of trigger status, **Zabbix server** has **no active problems**, while **Client Device** currently has **two active problem alerts** triggered by low free memory and high memory utilization.

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5) Where can you view the detailed status of all monitored items for a specific host?

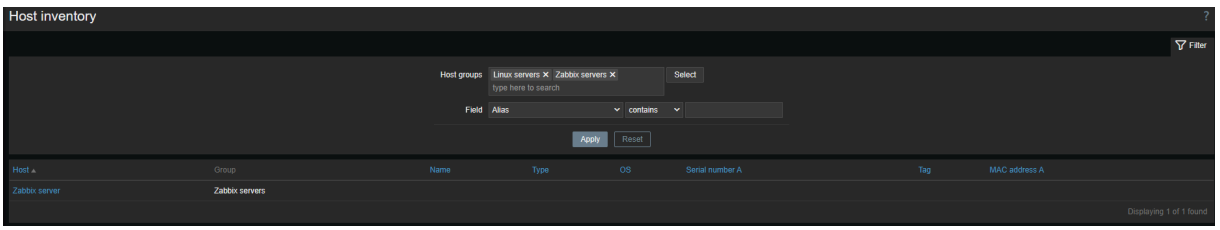
-> Navigate to Monitoring -> Latest data, select the host in the filter bar, and click Apply. Alternatively, navigate to Data collection -> Hosts and click the Items hyperlink on the target host row.

6) What is the significance of the latest data for your local host's HTTP service? Provide a screenshot and explain.



It confirms whether the web service process (Apache/Nginx) is actively listening, accepting TCP connections, and responding to client requests in real time. A returned value of 1 indicates the web daemon is operational and responsive, while 0 indicates connection refusal, application crash, or port failure, allowing monitoring independent of the OS host status.

7) From the Host Inventory, what are two types of information available about your monitored hosts?



1. **Operating System Details (OS): Displays the detected or recorded operating system and kernel version running on the host (e.g., Linux distribution details).**
2. **Hardware and Network Identifiers (MAC address A / Serial number A / Type): Tracks physical hardware attributes and device inventory, including network interface MAC addresses, equipment serial numbers, and device classifications.**

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